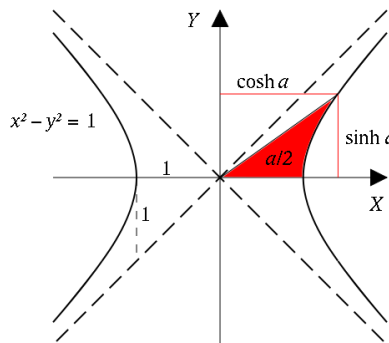


1.8 Hyperbolic Trigonometric Functions

Motivation: The trigonometric functions work well when we are trying to do trigonometry on a plane (intuitively a plane is a flat, infinite two dimensional object).

However, it doesn't always make sense to do trigonometry on the plane. For example, the earth is roughly a sphere and trigonometry on a sphere is different than trigonometry on a plane. Another example where trigonometry will be different is if we are on a hyperbolic plane (*the instructor will show an example of what a hyperbolic plane might look like*).

In hyperbolic trigonometry, for any real number t , the point $(\cosh(t), \sinh(t))$ ($\cosh(t)$ is read “hyperbolic cosine of t ” and $\sinh(t)$ is read “hyperbolic sine of t ”) lies on the right branch of the hyperbola $x^2 - y^2 = 1$ and is such that the signed area of the “sector” formed by the segment from the origin to $(1,0)$, the segment from the origin to $(\cosh(t), \sinh(t))$, and the arc of the hyperbola is: $t/2$. The image below from Wikipedia illustrates this idea:



It can be proven using the definition above and ideas from the start of Chapter 2 that for any real number x ,

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh(x) = \frac{e^x - e^{-x}}{2}.$$

Note: This result makes sense since the x coordinates are all positive on the right branch of the hyperbola above and y values may be both positive and negative.

This leads to the following definitions.

Definitions: We define six hyperbolic trig functions.

Pronounced "sinch" → $\sinh(x) = \frac{e^x - e^{-x}}{2}$

$$\operatorname{csch}(x) = \frac{1}{\sinh(x)}$$

Pronounced "cosh" → $\cosh(x) = \frac{e^x + e^{-x}}{2}$

$$\operatorname{sech}(x) = \frac{1}{\cosh(x)}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

$$\operatorname{coth}(x) = \frac{1}{\tanh(x)}$$

Example 1: Evaluate each of the following using the definition.

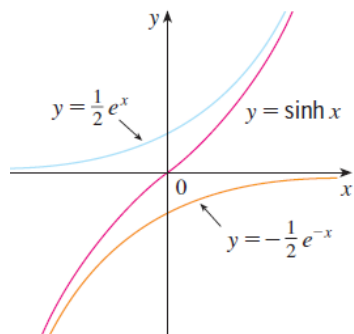
$$\text{a) } \sinh(0) = \frac{e^0 - e^0}{2} = \frac{0}{2} = 0$$

$$\text{b) } \cosh(1) = \frac{e^1 + e^{-1}}{2} = \frac{1}{2} \left(e + \frac{1}{e} \right)$$

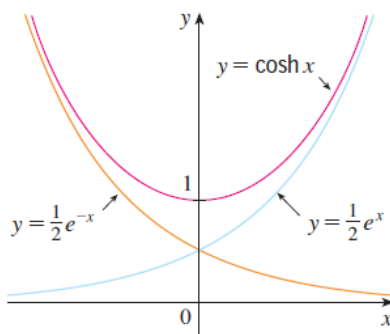
$$\text{c) } \tanh(2) = \frac{\left(\frac{e^2 - e^{-2}}{2} \right)}{\left(\frac{e^2 + e^{-2}}{2} \right)} = \frac{e^2 - e^{-2}}{e^2 + e^{-2}}$$

Graphs of hyperbolic functions:

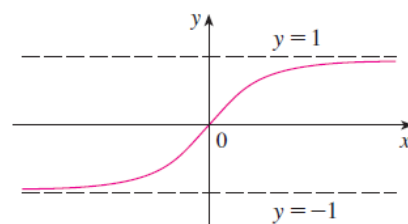
$$y = \sinh(x)$$



$$y = \cosh(x)$$



$$y = \tanh(x)$$



Theorem: We have the following identities for all real x that are in the domain of the functions involved in each identity:

- 1) $\sinh(-x) = -\sinh(x)$
- 2) $\cosh(-x) = \cosh(x)$
- 3) $\cosh^2(x) - \sinh^2(x) = 1$
- 4) $1 - \tanh^2(x) = \operatorname{sech}^2(x)$
- 5) $\coth^2(x) - 1 = \operatorname{csch}^2(x)$

Example 2: Show that $\cosh^2(x) - \sinh^2(x) = 1$.

$$\begin{aligned}\cosh^2(x) - \sinh^2(x) &= \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 \\&= \frac{e^{2x} + 2e^x e^{-x} + e^{-2x}}{4} - \frac{e^{2x} - 2e^x e^{-x} + e^{-2x}}{4} \\&= \frac{4}{4} = 1\end{aligned}$$

Example 3: Find $\frac{d}{dx}[\sinh(x)]$ and $\frac{d}{dx}[\cosh(x)]$.

$$\frac{d}{dx}[\sinh(x)] = \frac{d}{dx}\left[\frac{e^x - e^{-x}}{2}\right] = \frac{1}{2} \frac{d}{dx}[e^x - e^{-x}] = \frac{1}{2}[e^x + e^{-x}] = \frac{e^x + e^{-x}}{2} = \cosh(x)$$

&

$$\frac{d}{dx}[\cosh(x)] = \frac{d}{dx}\left[\frac{e^x + e^{-x}}{2}\right] = \frac{1}{2} \frac{d}{dx}[e^x + e^{-x}] = \frac{1}{2}[e^x - e^{-x}] = \frac{e^x - e^{-x}}{2} = \sinh(x)$$

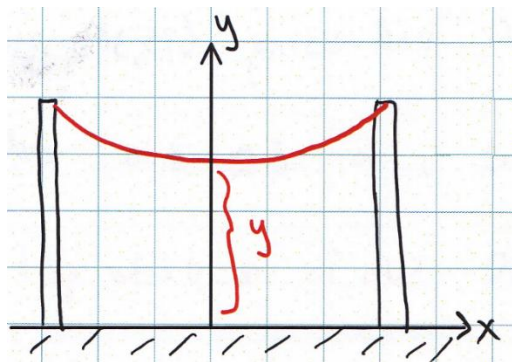
Theorem: We have the following derivatives for our six hyperbolic trig. functions.

- $\frac{d}{dx}[\sinh(x)] = \cosh(x)$
- $\frac{d}{dx}[\cosh(x)] = \sinh(x)$
- $\frac{d}{dx}[\tanh(x)] = \text{sech}^2(x)$
- $\frac{d}{dx}[\coth(x)] = -\text{csch}^2(x)$
- $\frac{d}{dx}[\text{sech}(x)] = -\text{sech}(x)\tanh(x)$
- $\frac{d}{dx}[\text{csch}(x)] = -\text{csch}(x)\coth(x)$

Applications

Hyperbolic functions arise when a homogeneous cable is suspended between two points.

Catenary Curve



$$y = a \cosh\left(\frac{x}{a}\right) + C$$

(where a and c are constants driven by the type of cable and the distance between the poles.)

Example 4: Suppose a cable is attached between two poles which stand 30ft apart and can be modelled by the function $y = \cosh(x) + 5$. Determine the exact length of the cable from $0 \leq x \leq 10$.

$y' = \sinh(x)$ is continuous on $[0, 10]$. Therefore the arc length of the cable is given by:

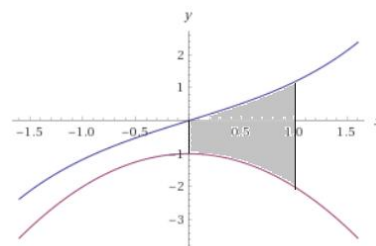
$$L = \int_0^{10} \sqrt{1 + \sinh^2(x)} dx = \int_0^{10} \sqrt{\cosh^2(x)} dx = \int_0^{10} |\cosh(x)| dx. \text{ Since } y = \cosh(x) \text{ is always positive we can drop the absolute value bars. Therefore,}$$

$$L = \int_0^{10} |\cosh(x)| dx = \int_0^{10} \cosh(x) dx = \sinh(x) \Big|_0^{10} = \sinh(10) - \sinh(0) = \sinh(10)$$

Example 5: Consider the region bounded by the curves $y = \sinh(x)$ & $y = -x^2 - 1$ from $0 \leq x \leq 1$.

a) Determine the exact value of the area of this region.

$$\begin{aligned} A &= \int_0^1 [\sinh(x) - (-x^2 - 1)] dx = \int_0^1 [\sinh(x) + x^2 + 1] dx \\ &= \left[\cosh(x) + \frac{1}{3}x^3 + x \right]_0^1 = \left[\cosh(1) + \frac{1}{3} + 1 \right] - [\cosh(0) + 0 + 0] \\ &= \cosh(1) + \frac{1}{3} + 1 - 1 \\ &= \cosh(1) + \frac{1}{3} \end{aligned}$$



- b) Set up, *but do not evaluate*, and integral which represents the volume when this region is revolved around the y -axis.

This question is simple to answer if we do an integral with respect to x . This will require the use of cylindrical shells.

$$V = \int_0^1 2\pi x [\sinh(x) - (-x^2 - 1)] dx$$

OR

If we choose to use washers we have to calculate integrals with respect to y . We would obtain.

$$V = \int_{-2}^{-1} [\pi(1)^2 - \pi(\sqrt{-1-y})^2] dy + \int_{-1}^0 \pi(1)^2 dy + \int_0^{\sinh(1)} [\pi(1)^2 - \pi(\sinh^{-1}(y))^2] dy$$

- c) Suppose you rotate the curve $y = \sinh(x)$ on $-1 \leq x \leq 1$ about the x -axis and attempt to determine the surface area of the resulting solid. Why does the integral $\int_{-1}^1 2\pi \sinh(x) \sqrt{1 + \cosh^2(x)} dx$ *not* represent this surface area?

The reason that this integral does not compute the desired surface area is because the function $y = \sinh(x)$ is negative for $x < 0$ and therefore it does not meet the conditions for the use of the surface area integral.

